 

# KOZ'MODEM'YANSK, Russian Federation

WMO: **274790**

Lat: **56.333N** Lon: **46.583E** Elev: **352** StdP: **14.51** Time Zone: **3.00 (E03)** Period: **94-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB 99.6% | Heating DB 99% | Humidification 99.6% DP | Humidification 99.6% HR | Humidification 99.6% MCDB | Humidification 99% DP | Humidification 99% HR | Humidification 99% MCDB | Coldest Month 0.4% WS | Coldest Month 0.4% MCDB | Coldest Month 1% WS | Coldest Month 1% MCDB | MCWS to 99.6% DB | PCWD to 99.6% DB | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | 1 | -15.3 | -9.5 | -20.6 | 1.8 | -14.8 | -15.0 | 2.5 | -8.7 | 18.8 | 20.1 | 17.6 | 18.7 | 3.6 | 320 | 0.531 | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling 0.4% DB | Cooling 0.4% MCWB | Cooling 1% DB | Cooling 1% MCWB | Cooling 2% DB | Cooling 2% MCWB | Evaporation 0.4% WB | Evaporation 0.4% MCDB | Evaporation 1% WB | Evaporation 1% MCDB | Evaporation 2% WB | Evaporation 2% MCDB | MCWS to 0.4% DB | PCWD to 0.4% DB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | 7 | 13.3 | 85.3 | 68.7 | 82.2 | 68.0 | 79.3 | 66.5 | 71.2 | 81.4 | 69.6 | 79.3 | 68.0 | 76.9 | 5.0 | 250 | (2) |

| | Dehumidification 0.4% DP | Dehumidification 0.4% HR | Dehumidification 0.4% MCDB | Dehumidification 1% DP | Dehumidification 1% HR | Dehumidification 1% MCDB | Dehumidification 2% DP | Dehumidification 2% HR | Dehumidification 2% MCDB | Enthalpy 0.4% Enth | Enthalpy 0.4% MCDB | Enthalpy 1% Enth | Enthalpy 1% MCDB | Enthalpy 2% Enth | Enthalpy 2% MCDB | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | 67.6 | 102.9 | 76.4 | 66.1 | 97.4 | 74.8 | 64.6 | 92.2 | 73.5 | 35.2 | 81.6 | 33.8 | 79.2 | 32.5 | 77.0 | 80.2 | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS 1% | Extreme Annual WS 2.5% | Extreme Annual WS 5% | | Extreme Annual Temperature Mean Min | Mean Max | Standard Deviation Min | Standard Deviation Max | n=5 years Min | n=5 years Max | n=10 years Min | n=10 years Max | n=20 years Min | n=20 years Max | n=50 years Min | n=50 years Max | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | 16.8 | 15.0 | 13.5 | DB | -19.6 | 89.1 | 6.6 | 3.4 | -24.4 | 91.6 | -28.3 | 93.6 | -32.0 | 95.5 | -36.8 | 98.0 | (4) |
| (5) | | | | WB | -20.0 | 73.6 | 6.5 | 2.3 | -24.6 | 75.3 | -28.5 | 76.6 | -32.1 | 77.9 | -36.9 | 79.6 | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | | DBAvg | 41.6 | 14.5 | 15.6 | 26.7 | 42.6 | 56.9 | 64.3 | 68.8 | 64.8 | 54.4 | 41.6 | 28.4 | 18.9 | (6) |
| (7) | | | DBStd | 21.55 | 12.64 | 12.35 | 8.15 | 9.18 | 9.29 | 7.86 | 6.63 | 7.36 | 7.72 | 7.98 | 9.85 | 12.13 | (7) |
| (8) | | | HDD50 | 5018 | 1101 | 963 | 721 | 259 | 39 | 3 | 0 | 0 | 41 | 281 | 647 | 963 | (8) |
| (9) | | | HDD65 | 8909 | 1566 | 1383 | 1186 | 673 | 282 | 107 | 37 | 94 | 329 | 727 | 1097 | 1428 | (9) |
| (10) | | | CDD50 | 1954 | 0 | 0 | 0 | 36 | 253 | 432 | 582 | 460 | 172 | 19 | 0 | 0 | (10) |
| (11) | | | CDD65 | 370 | 0 | 0 | 0 | 0 | 31 | 86 | 154 | 89 | 10 | 0 | 0 | 0 | (11) |
| (12) | | | CDH74 | 2266 | 0 | 0 | 0 | 1 | 178 | 548 | 985 | 524 | 30 | 0 | 0 | 0 | (12) |
| (13) | | | CDH80 | 546 | 0 | 0 | 0 | 0 | 29 | 127 | 243 | 144 | 3 | 0 | 0 | 0 | (13) |
| (14) | Wind | | WSAvg | 6.5 | 6.9 | 6.9 | 6.9 | 6.3 | 6.4 | 5.7 | 5.0 | 5.4 | 6.0 | 7.2 | 7.3 | 7.3 | (14) |
| (15) | Precipitation | | PrecAvg | 23.8 | 1.7 | 1.5 | 1.5 | 1.3 | 1.6 | 2.7 | 2.8 | 2.3 | 2.2 | 2.1 | 2.0 | 1.9 | (15) |
| (16) | | | PrecMax | 29.5 | 3.7 | 2.9 | 2.8 | 3.2 | 2.6 | 6.5 | 5.7 | 4.8 | 4.5 | 4.1 | 3.5 | 3.5 | (16) |
| (17) | | | PrecMin | 17.0 | 0.7 | 0.6 | 0.2 | 0.3 | 0.6 | 0.6 | 0.5 | 0.0 | 0.5 | 0.1 | 0.6 | 0.7 | (17) |
| (18) | | | PrecStd | 2.8 | 0.6 | 0.6 | 0.7 | 0.8 | 0.6 | 1.5 | 1.4 | 1.3 | 1.1 | 1.1 | 0.7 | 0.6 | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | 36.5 | 37.1 | 48.3 | 71.2 | 83.6 | 88.1 | 90.8 | 90.3 | 77.8 | 64.7 | 51.2 | 41.9 | (19) |
| (20) | | | MCWB | | 34.8 | 34.1 | 41.0 | 55.3 | 65.3 | 69.3 | 69.8 | 70.1 | 62.4 | 54.4 | 47.9 | 40.2 | (20) |
| (21) | | 2% | DB | | 34.6 | 35.4 | 42.7 | 65.7 | 78.9 | 83.5 | 85.4 | 83.8 | 73.0 | 59.4 | 46.0 | 37.7 | (21) |
| (22) | | | MCWB | | 33.2 | 33.5 | 36.7 | 51.6 | 62.6 | 68.0 | 69.2 | 68.9 | 61.0 | 52.0 | 43.6 | 36.1 | (22) |
| (23) | | 5% | DB | | 33.2 | 33.9 | 39.3 | 61.4 | 74.8 | 80.2 | 82.3 | 79.2 | 69.4 | 55.5 | 42.7 | 35.2 | (23) |
| (24) | | | MCWB | | 32.1 | 32.6 | 34.7 | 49.8 | 60.4 | 66.7 | 68.8 | 67.0 | 59.4 | 50.1 | 40.8 | 33.8 | (24) |
| (25) | | 10% | DB | | 31.4 | 32.2 | 36.9 | 57.0 | 70.9 | 76.7 | 79.6 | 76.0 | 65.7 | 52.5 | 39.4 | 33.6 | (25) |
| (26) | | | MCWB | | 30.3 | 31.0 | 33.7 | 47.5 | 58.0 | 64.5 | 67.5 | 65.4 | 57.6 | 48.5 | 37.8 | 32.4 | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | 35.1 | 35.2 | 42.0 | 56.1 | 67.3 | 72.3 | 73.8 | 72.5 | 64.8 | 56.1 | 47.9 | 40.5 | (27) |
| (28) | | | MCDB | | 35.9 | 36.1 | 48.0 | 67.5 | 80.7 | 83.4 | 83.6 | 83.8 | 73.2 | 61.1 | 50.0 | 41.4 | (28) |
| (29) | | 2% | WB | | 33.6 | 33.8 | 37.5 | 53.2 | 64.3 | 69.9 | 71.5 | 70.5 | 62.3 | 53.1 | 44.2 | 36.3 | (29) |
| (30) | | | MCDB | | 34.4 | 34.7 | 41.1 | 62.5 | 76.4 | 80.1 | 81.4 | 80.8 | 70.5 | 57.6 | 46.1 | 37.3 | (30) |
| (31) | | 5% | WB | | 32.2 | 32.7 | 35.7 | 50.8 | 61.8 | 68.0 | 70.1 | 68.5 | 60.2 | 50.7 | 40.9 | 34.1 | (31) |
| (32) | | | MCDB | | 33.2 | 33.9 | 38.4 | 60.5 | 72.3 | 77.2 | 79.6 | 77.4 | 67.4 | 54.7 | 42.5 | 34.9 | (32) |
| (33) | | 10% | WB | | 30.2 | 31.0 | 34.0 | 47.8 | 59.7 | 66.0 | 68.6 | 66.5 | 58.3 | 48.7 | 37.9 | 32.5 | (33) |
| (34) | | | MCDB | | 31.4 | 32.4 | 36.4 | 56.4 | 69.2 | 75.5 | 77.6 | 74.1 | 64.6 | 52.2 | 39.3 | 33.5 | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | 9.2 | 10.4 | 10.6 | 13.1 | 14.7 | 13.7 | 13.3 | 13.0 | 11.0 | 8.1 | 6.7 | 8.0 | (35) |
| (36) | | 5% DB | MCDBR | | 7.1 | 7.8 | 11.6 | 18.9 | 18.4 | 16.9 | 16.3 | 16.6 | 15.8 | 12.2 | 8.6 | 6.7 | (36) |
| (37) | | | MCWBR | | 7.1 | 7.4 | 8.1 | 9.9 | 8.9 | 7.9 | 7.2 | 7.3 | 8.1 | 7.9 | 7.8 | 6.2 | (37) |
| (38) | | 5% WB | MCDBR | | 7.1 | 7.5 | 10.2 | 18.0 | 17.2 | 15.5 | 14.3 | 15.2 | 14.1 | 11.2 | 8.1 | 6.6 | (38) |
| (39) | | | MCWBR | | 7.0 | 7.3 | 8.0 | 10.2 | 9.1 | 8.1 | 7.4 | 7.6 | 8.2 | 8.3 | 7.8 | 6.6 | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | 0.286 | 0.318 | 0.355 | 0.404 | 0.381 | 0.376 | 0.404 | 0.425 | 0.371 | 0.350 | 0.313 | 0.282 | (40) |
| (41) | | taud | | | 2.078 | 2.059 | 2.031 | 2.143 | 2.356 | 2.406 | 2.335 | 2.253 | 2.389 | 2.338 | 2.214 | 2.059 | (41) |
| (42) | | Ebn at Noon | | | 183 | 226 | 248 | 256 | 272 | 275 | 263 | 247 | 247 | 218 | 175 | 149 | (42) |
| (43) | | Edh at Noon | | | 19 | 30 | 40 | 42 | 36 | 35 | 36 | 37 | 28 | 23 | 17 | 16 | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | 180 | 455 | 925 | 1289 | 1725 | 1839 | 1813 | 1408 | 854 | 390 | 153 | 99 | (44) |
| (45) | | RadStd | | | 28 | 90 | 105 | 131 | 118 | 147 | 162 | 140 | 103 | 61 | 27 | 24 | (45) |

## Historical Trends

| | | DBAvg | Heating 99% DB | Heating 99% DP | Cooling 1% DB | Cooling 1% WB | Cooling 1% DP | Degree-Days HDD50 | HDD65 | CDD50 | CDD65 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | +1.10 | N/A | N/A | N/A | N/A | N/A | -274 | -355 | N/A | N/A | (46) |
| (47) | Regional (8 neighbors) | +1.01 | N/A | N/A | N/A | N/A | N/A | -238 | -320 | N/A | N/A | (47) |

Nomenclature: See separate page